

ASSIGNMENT - 10**Complex form of Fourier series**

1. Find the complex form of the Fourier series of $f(x) = \cosh 3x + \sinh 3x$ in $(-3, 3)$

$$\text{Ans: } \sinh 9 \sum_{n=-\infty}^{\infty} \frac{(-1)^n (9 + n\pi)}{(81 + (n\pi)^2)} e^{\frac{in\pi x}{3}}$$

2. Find the complex form of the Fourier series of the periodic function $f(x) = \begin{cases} -k & \text{in } (-\pi, 0) \\ k & \text{in } (0, \pi) \end{cases}$

$$\text{Ans: } f(x) = \frac{k}{i\pi} \sum_{n=-\infty}^{\infty} \frac{(1 - (-1)^n)}{n} e^{inx}$$